INCH-POUND

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SUPERSEDING
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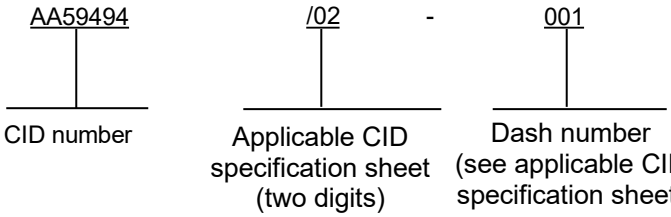
COMMERCIAL ITEM DESCRIPTION

WHEELS AND HUBS FOR INDUSTRIAL PNEUMATIC TIRES

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the general requirements for wheels and hubs, with or without industrial pneumatic tires and tubes. Requirements for specific wheel and hub assemblies are covered in the individual CID specification sheets. Wheels and hubs, with or without industrial pneumatic tires and tubes covered by this CID are intended for commercial/industrial applications.

2. CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This CID uses a classification system which is included in the PIN as shown in the following example (see 7.1).



3. SALIENT CHARACTERISTICS.

3.1 Interface and physical dimensions. Wheels, hubs, and tires supplied to this CID shall be as specified on the applicable CID specification sheet.

3.2 CID specification sheet. The family of wheels, hubs, and tires shall be in accordance with the requirements specified herein and the applicable CID specification sheet. In the event of a conflict between this general CID and the applicable CID specification sheet, the latter shall govern.

3.3 Performance.

3.3.1 Pressure loads. Wheels, when mounted with tires and tubes, shall not deform when the tires and tubes are subjected to the inflation pressure specified in table I.

Comments, suggestions, or questions on this document should be addressed to: DLA Land and Maritime, Attn: VAI, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to FluidFlow@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

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TABLE I. Minimum wheel deformation inflation pressure.

Size	Ply rating	Inflation pressure, psi
3.50-6	4-ply	140
4.00-8	6-ply	175
6.00-9	10-ply	175
7.50-10	10-ply	175
9.00-10	10-ply	175

3.3.2 Static loads. Wheels, when mounted with tires and tubes and mounted to applicable hubs, shall withstand the maximum static loads specified in table II.

TABLE II. Wheel maximum static loads.

Wheel size	Tire inflation, psi	Maximum static load, lbs.(at each wheel)
3.50-6	80	2,500
4.00-8	100	5,000
6.00-9	100	6,250
7.50-10	95	10,000
9.00-10	80	10,000

3.3.3 Towing. Wheel and hub assemblies shall be capable of being moved by hand or towed by a tug over concrete pavement in a circular path having a 35 foot radius (as measured to the outside of the wheels) for a minimum of one-half mile at speeds up to 15 miles per hour (mph) under the 10 percent overrated load condition for 5 mph specified in tables III and IV.

3.4 Usage. The wheels and hubs shall be capable of being used as running gear on equipment that may be moved by hand or towed by a tug on paved areas, pierced plank areas, unpaved roads, and unimproved level terrain, whether such areas are dry, wet, or covered with up to 6 inches of snow or 4 inches of mud.

3.5 Design and construction. Wheel and hub assemblies shall be constructed so that no parts will work loose in service.

3.5.1 Materials. Wheels and hubs shall be made out of steel, iron, or aluminum alloy as specified in their respective specification sheet. These materials, which are subject to deterioration when exposed to climatic and environmental conditions, shall be of corrosion-resistant type metal or shall be of metal that is protected against deterioration in a manner that will in no way prevent compliance with the requirements of this CID. The use of any protective coating that will crack, chip, or scale with age or extremes of climate and environmental conditions shall be avoided.

3.5.2 Wheels. Wheels shall be of the split-rim type, with all of the necessary bolts, lockwashers, and nuts required for assembly of the two halves. The wheels shall allow removal of the wheel from the hub without deflating the tire.

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TABLE III. Industrial tires, loads, and inflation pressure for paved roads.[1]

Tire Size (TRA)	Ply Rating	Valve hole location	TRA rim contour number	Tire OD (inches) (approx.)	On paved road					
					5 MPH		10 MPH		50 MPH	
					Load rating (lbs)[2]	Inflation pressure (psi)	Load rating (lbs)[2]	Inflation pressure (psi)	Load rating (lbs)[2]	Inflation pressure (psi)
3.50-6	4	Center	3-1/4 I	13	590	80	490	75	295	65
4.00-8	6	Center	3-3/4 I	16	1,090	100	950	100	570	85
6.00-9	10	9/16-inch offset	4.00 E	21	2,290	100	1,990	100	1,175	95
7.50-10	10	1-inch offset	5.50 F	25	3,520	95	2,860	85	1,620	75
9.00-10[4]	10	1-inch offset	5.50 F	28	4,270	80	3,580	75	2,000	65

NOTES: See “NOTES” for Table IV

TABLE IV. Industrial tires, loads, and inflation pressure for off-roads.[1]

Tire Size (TRA)	Ply Rating	Valve hole location	TRA rim contour number	Tire OD (inches) (approx.)	Off-road[3]	
					10 MPH	
					Load rating (lbs)[2]	Inflation pressure (psi)
3.50-6	4	Center	3-1/4 I	13	220	30
4.00-8	6	Center	3-3/4 I	16	370	30
6.00-9	10	9/16-inch offset	4.00 E	21	780	30
7.50-10	10	1-inch offset	5.50 F	25	1,230	30
9.00-10[4]	10	1-inch offset	5.50 F	28	1,655	30

NOTES:

1. Table I is based on the recommendations of The Tire and Rim Association, Inc (TRA). Worst case conditions of use, not average conditions, should be used in tire selection. For given conditions of use, loads below the load rating specified permit use of lower inflation pressures and hence affords an increase in flotation.
2. Loads specified apply to both ferrous and aluminum alloy wheels.
3. No tire smaller than size 6.00-9 should be used as the main wheels for equipment to be used in unpaved areas. The use of sizes 3.50-6 and 4.00-8 should be limited to auxiliary wheels.
4. Ferrous wheels and hubs only.

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3.5.3 Hubs. A hub shall consist of all the parts shown on the respective specification sheet for the hub assembly, as applicable, including all bolts, nuts, and lockwashers for attaching the wheel to the hub. All parts such as grease seals and hubcaps that are not specified in detail shall meet the requirements specified in 3.6. Axles are not a part of the hubs covered by this CID.

3.5.3.1 Bearing cup removal. Slots or other provisions shall be inside the hubs to facilitate removal of bearing cups.

3.5.3.2 Attachment bolts. Hub-to-wheel attachment bolts shall be of the sizes shown on the applicable specification sheet and shall be ribbed-neck carriage bolts. The length of the ribbed part of the bolt shall be from the head to within 1/16 inch of extending through the flange of the hub. Bolts shall be capable of being pressed out to allow for attachment of a brake mechanism to the hub body.

3.5.3.3 Seals. Seals for the hub as shown on the respective specification sheet for the hub assembly shall be standard grease seals of the molded, synthetic-rubber type. The seals shall provide protection for the wheel bearings under the environmental conditions specified herein.

3.5.3.4 Brake drum pilot. Each type 1 cantilever mounting hub shall be provided with a brake drum pilot surface on the hub flange.

3.5.4 Threads. Unless other specified, all threads shall conform to FED-STD-H28 (see 7.4).

3.6 Environmental conditions. The wheels and hubs shall be capable of withstanding exposure to the following environmental conditions:

a. Temperatures from -65 to 125 □F.

b. Relative humidity of 95□5 percent at a temperature of 125□5 □F.

c. Airborne sand and dust particles incident to normal road operation.

d. Atmosphere containing salt-laden moisture (a salt fog of 5□1 percent salt solution).

3.7 Tires and tubes. All industrial pneumatic-type tires and tubes furnished with the wheels shall conform to ZZ-T-410, except that tire size 7.50-10 shall be 10-ply rating. Tire size 9.00-10 shall be used only with ferrous metal hubs and wheels. All tires shall have a free-rolling tread design, be of a non-directional type, and have multiple ribs.

3.7.1 Flaps. Unless otherwise specified (see 7.5), all 10-ply rating tires shall be equipped with flaps.

3.7.2 Valve stem. The valve stem on the tube furnished with the tire shall conform to the appropriate TRA Standard (see 7.4) as indicated:

Tire size	TRA Standard
3.50-6	TR 87
4.00-8	TR 13 or TR 87
6.00-9 and 7.50-10	TR 135
9.00-10	TR 150

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3.8 Finishes and protective coatings.

3.8.1 Color. Wheels and hubs shall be finished in the color specified in the contract or order (see 7.5).

3.8.2 Bearings, grease seals, enclosed parts, and utility parts. The bearings, grease seals, and enclosed parts and surfaces of the hub shall not be painted. The outside surface of the seals shall be painted or plated in accordance with the applicable finishes specified above. Bearing cup bores shall not be finished. Utility parts (e.g., wheel mounting bolts) shall be masked or otherwise protected during painting of the hubs.

3.9 Marking. Wheels and hubs supplied to this CID shall be marked with the manufacturer's (MFR's) standard commercial PIN. (NOTE: The part number marked on the unit pack shall be the CID PIN.)

3.10 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.11 Workmanship. The wheels and hubs shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

3.11.1 Wheel surface. The surface of the wheel between bead seats shall be free of defects that would be injurious to the innertube and tire.

3.11.2 Flanges. The hub face and mating wheel faces shall be smooth and free from defects.

3.11.3 Register. The register bore of the wheel and register diameter of the hub shall be concentric and free from defects. The least clearance between these surfaces shall be maintained consistent with the applicable figures.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptance. The following market acceptance criteria are necessary to document the quality of the product to be provided under this CID:

a. The company producing the item must have been producing a product meeting the requirements of this CID for at least 1 year.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 PIN. The PIN should be used for Government purposes to buy commercial products to this CID. See section 2 for PIN format example.

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7.2 Environmentally preferable material. Environmentally preferable materials should be used to the maximum extent possible to meet the requirements of this specification. As of the dating of this document, the U.S. Environmental Protection Agency (EPA) is focusing efforts on reducing 31 priority chemicals. The list of chemicals and additional information is available on their website https://www.epa.gov/osw/hazard/wastemin/priority.htm. Included in the EPA list of 31 priority chemicals are cadmium, lead, and mercury. Use of these materials should be minimized or eliminated unless needed to meet the requirements specified herein (see Section 3).

7.3 Commercial and Government Entity (CAGE) code. For ordering purposes, inventory control, and submission of these wheels and hubs to DSCC under the Military Parts Control Advisory Group (MPCAG) evaluation program, CAGE code 58536 should be used.

7.4 Source of documents.

Federal Specification

ZZ-T-410	-	Tires, Pneumatic, Industrial

Federal Standards

FED-STD-H28	-	Screw-Thread Standards for Federal Services

Commercial Item Descriptions

A-A-59494/1	-	Wheels and Hub Assemblies
A-A-59494/2	-	Wheel, Pneumatic Tire, 3.50-6
A-A-59494/3	-	Wheel, Pneumatic Tire, 4.00-8
A-A-59494/4	-	Wheel, Pneumatic Tire, 6.00-9
A-A-59494/5	-	Wheel, Pneumatic Tire, 7.50-10 or 9.00-10
A-A-59494/6	-	Hub Assembly, Cantilever Mounting, for 3.50-6 and 4.00-8
A-A-59494/7	-	Hub Assembly, Cantilever Mounting, for 6.00-9, 7.50-10, and 9.00-10
A-A-59494/8	-	Hub Assembly, Fork Mounting, for 3.50-6 and 4.00-8 Wheels
A-A-59494/9	-	Hub Assembly, Fork Mounting, for 6.00-9 and 7.50-10 Wheels

(Copies of these documents are available online at https://assist.dla.mil)

Federal Regulations

FAR	-	Federal Acquisition Regulation (FAR)

(Copies of these documents are available online at https://www.acquisition.gov/Far)

Other Publications

SAE INTERNATIONAL

SAE-AMS-STD-595	-	Colors Used in Government Procurement

(Copies of these documents are available online at https://www.sae.org/ from SAE International, or email at CustomerService@sae.org.)

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Tire and Rim Association (TRA)

The Tire and Rim Association (TRA) Standards

(Copies of these documents may be obtained online at https://www.us-tra.org/traPubs.html)

Aerospace Industries Association of America

NASM24665 - Pin, Cotter (Split)

(Copies of these documents may be obtained online at https://global.ihs.com/home_page_aia.cfm?&csf=AIA)

7.5 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN

b. Product conformance provisions.

c. Packaging requirements.

d. Color code designation per SAE-AMS-STD-595.

7.6 Government users. To acquire information on obtaining these rubber steam hoses or hose assemblies from the Government inventory system, contact DLA Customer Interaction Center (CIC), email dlacontactcenter@dla.mil, or toll free 1-877-DLA-CALL (1-877-352-2255), DSN CONUS/OCONUS: 877-352-2255, Commercial: 269-704-7921, or Fax: 269-704-7930.

7.7 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

MILITARY INTERESTS:

Custodians:
Army - AT
Air Force -184
DLA – CC

Review activity:
Air Force - 84

CIVIL AGENCY AND COORDINATING ACTIVITY:

GSA - FAS

Preparing activity:
DLA - CC

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